

Identification and Cholesky

Status: draft for review

Question and intuition

Reduced-form residuals mix structural shocks. Identification asks which economic disturbance generated them. A covariance matrix alone generally cannot distinguish monetary policy, demand, supply and information shocks. Timing assumptions add restrictions that must be defended.

Model and assumptions

$u_t = B\varepsilon_t$, $E[\varepsilon_t\varepsilon_t'] = I$, hence $\Sigma = BB'$. A lower-triangular Cholesky factor chooses one of many admissible decompositions. Ordering a variable before policy means its impact response to that policy shock is zero.

Interpretation and common errors

Normalize the shock explicitly: a one-unit structural innovation is not a 100 bp tightening unless the policy impact has been rescaled. With monthly averages, the central bank may react inside the same month to exchange-rate movements. A policy-before-FX ordering can therefore recover a contemporaneous response to depreciation rather than an exogenous tightening. Swapping the ordering changes identifying restrictions; it is not a harmless relabelling.

Portfolio application

EA two-year yields reflect expectations and information as well as policy actions. High-frequency surprises add an external instrument if relevant and exogenous to non-policy structural shocks. A strong first-stage F addresses relevance only. Information contamination, event-window choice and sample alignment remain substantive issues. Sign restrictions identify a set, and medians across rotations need not correspond to one admissible model.

Exam practice

1. Does Cholesky prove exogeneity?

Answer. No. It imposes recursive timing restrictions.

2. What changes when FX precedes policy?

Answer. Policy cannot move FX on impact; policy may react to FX contemporaneously.

3. Why normalize a tightening?

Answer. To compare responses to a common policy-indicator impact rather than shock variance.

4. What does a strong first stage establish?

Answer. Instrument relevance under the reported test, not exclusion.

5. Why label sign spread differently from a CI?

Answer. Rotation variation is identification uncertainty conditional on the reduced form, not sampling coverage.

Worked reasoning

Before interpreting an estimate, name the outcome unit, shock unit, identifying restrictions, sample and uncertainty method. If any is missing, the numerical sign alone does not support a causal claim. Separate a mechanical accounting identity from a maintained economic restriction. Explain how the answer would change under a different permissible identification without searching until a preferred answer appears. A correct exam response states what can and cannot be inferred, rather than merely naming a test or estimator.

Reference and further reading

Verified primary reference: <https://doi.org/10.1016/j.jmoneco.2019.08.016>

These notes are original explanations written for this portfolio; no thesis text or code was used.